We have all noticed that when you heat something up, its temperature rises. Often we think that heat and temperature are the same thing. However, this is not the case. Heat and temperature are related to each other, but are different concepts.

Heat is the total energy of molecular motion in a substance while temperature is a measure of the average energy of molecular motion in a substance. Heat energy depends on the speed of the particles, the number of particles (the size or mass), and the type of particles in an object. Temperature does not depend on the size or type of object. For example, the temperature of a small cup of water might be the same as the temperature of a large tub of water, but the tub of water has more heat because it has more water and thus more total thermal energy.

ACTIVITY:

Discuss what you think the answer is to the following question: Which object contains the most heat, a boiling pot of water or a gigantic iceberg?

Click here for the answer.

It is heat that will increase or decrease the temperature. If we add heat, the temperature will become higher. If we remove heat the temperature will become lower. Higher temperatures mean that the molecules are moving, vibrating and rotating with more energy.

If we take two objects which have the same temperature and bring them into contact, there will be no overall transfer of energy between them because the average energies of the particles in each object are the same. But if the temperature of one object is higher than that of the other object, there will be a transfer of energy from the hotter to the colder object until both objects reach the same temperature.

Temperature is not energy, but a measure of it. Heat is energy.

Pose the following question to your students.Ask students to justify their answers.

Which object contains the **most** heat, a boiling pot of water or a gigantic iceberg?

NOTES FOR TEACHERS:

This discussion is designed to help students differentiate between **heat** and **temperature** .

This question can also be turned around by asking students to consider a pot filled with ice and the iceberg. Assume they are at the same temperature,say -10°C, and you wish to melt each of them. Which of these objects would absorb the most heat to get to a temperature of +10°C ?
This should help them realize that the mass, not just temperature, is an important consideration in determining the heat (energy) content of an object.

This might be good opportunity to ask students if they think they are other properties of an object which might affect its heat content (as lead up to the concept of **specific heat**).

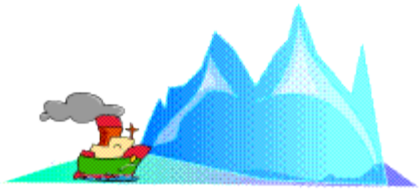
Consider the following...

SMALL HOT OBJECT



ahhh....steaming tea

LARGE COLD OBJECT



brrrr....iceberg

The total heat content in 500mL (0.5kg) of boiling water in a tea pot can be approximated by determining how much energy would be removed in order to cool it to absolute zero (about -273°C.)

Although the boiling water has a relatively high temperature, it contains relatively little heat energy.

Suppose a you were to cool a 1,000,000 tonne iceberg (10^{9}kg) to absolute zero. You could approximate the heat energy removed from the iceberg in order to do this.

While the iceberg is relatively cold (low temperature) it contains an enormous amount of heat energy relative to the boiling water in the tea pot.